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An Intelligent Approach for Semantic Plagiarism Detection in Scientific Papers

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***Abstract*— Plagiarism is the attempt to steal someone else's idea and portray it as your own, which is both unlawful and unethical. Plagiarism detection is described as the automatic detection of written items that have been reused. Modern Plagiarism detection software often examines text documents using a text representation and text similarity approaches to look for words that appear in both suspicious and source documents. The widespread use of the internet and simple access to textual content has increased the demand for automated plagiarism detection. This paper proposes an intelligent approach to detect semantic plagiarism in scientific papers. A corpus for storing the text of source scientific papers in order to compare the suspicious documents with it and detect plagiarism have been created. The documents are clustered into numerous groups using the Mini-Batch K-Means clustering algorithm, then get the documents in a specified category. The Universal Sentence Encoder (USE) intelligent model, was employed to extract textual features. The report of plagiarism percentage will be displayed as a web page and can be downloaded as a pdf file. A decision is made to accept or reject the suspicious scientific paper according to a threshold value. To evaluate the proposed system, it is compared with Turnitin platform for Plagiarism Detection (PD), the result of comparison indicates that the proposed system is reliable. The findings promote and can reduce the percentage of plagiarism after deleting some unnecessary parts of the suspicious document.**

Keywords*— *Plagiarism, Mini-Batch K-Means, USE, Cosine Similarity

I. INTRODUCTION

Plagiarism is defined as presenting someone else's work or ideas as one's own without explicitly acknowledging the original source, which is both immoral and illegal. The detection and prevention of such duplications is critical in this regard. Textual data must be mathematically represented after been processed through the techniques of Natural Language Processing (NLP) algorithms and hence help in such detections [1].

Text similarity measures the closeness between two texts. The detection of plagiarism, particularly in scientific documents, is a well-known use of text similarity concept [2],[3].

Intelligent algorithms in text mining applications represent words or sentences as a list of numbers[4]. This numerical representation includes a significant portion of the text data's synthetic and semantic rules[5].The usage of vectorized representations of text data allows for easier comparison of words and sentences while also reducing the requirement for lexicons [6].

Plagiarism can include alterations in the semantic representation of the text [2],[7]. If two texts have a same meaning, they are considered as semantically similar.

Extrinsic plagiarism detection aims to find plagiarism in a text through testing all of the available sources while Intrinsic plagiarism detection examines a questionable document and attempts to identify parts of it that were not created by the same author [8], [9].

Furthermore, Plagiarism is classified into three categories according to the number of hijacked words, phrases, or paragraphs: severe or entire plagiarism (more than 70%), intermediate plagiarism (between 30% and 70%), and light plagiarism (less than 30%)[10].

This paper's primary contributions are: i) Create a corpus consisting of 500 scientific papers, ii) Using a parser to exclude the unnecessary parts from the suspicious document and intelligent technique for clustering documents to check only related documents iii) finally generate a pretty plagiarism ratio report as a web page and as PDF file.

The paper is divided into seven sections. The second section goes into previous studies. The third section outlines the required methodologies in the field of Plagiarism Detection (PD). In section four, you will find further information on the proposed approach. The results of the experiments are provided in section five. Finally, parts six and seven presents the discussion, conclusion and suggestions for future works respectively.

II. LITERATURE REVIEW

A number of studies have been conducted on the use of information technology to detect plagiarism. Using the algorithms of cosine distance and K-means, W. Usino et al. [11] proposed a computerized approach for detecting information about plagiarism. The process starts with a novel step of checking an Indonesian large dictionary, followed by the building of a vector space model and a combined computation of K-means and cosine distance from 17 test items.

In I. Indriyanto and I. Sumitra [7], the cosine similarity is combined with the preprocessing techniques and TF-IDF to calculate the level of similarity between the title and abstract of a student's final scientific thesis. The results are then displayed and compared to the existing final project repository based on the threshold value to determine whether scientific work can be accepted or rejected.

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P. Hrkút, et. Al. [12] constructed a plagiarism detection method for a big dataset of source codes. The entire system is made up of various components. A classification of source code is an important aspect of the system since it divides data into individual clusters, making it easier to search vast amounts of source code. This is because grouping similar parts of the source code into appropriate clusters allows for a faster and more memory-efficient search for similar parts of the code.

The techniques of machine learning (ML) and (NLP) were used in the suggested system by F. K. AL-Jibory [13] , an external plagiarism detection approach based on text mining and similarity analysis was used by the system. The proposed method employs a mixture of Jaccard and cosine similarities.

Keya et al. [14] evaluated how closely the context and question-by-question system matched. Then they locate the response that is the closest to the original query. They measured the similarity using the cosine distance of the text and used USE for embedding. To discover the best related text, they employed deep averaging network (DAN).

H. El Mostafa and B. Faouzia [1] offered a deep Learning Algorithm-based method for detecting plagiarism in thoughts. The suggested method addresses some of the issues that can arise when attempting to detect plagiarism of ideas, such as loss of meaning or the difficulty in recognizing semantic similarities between documents. So, first, use doc2vec to produce a numerical representation of each sentence in a document, then use the Siamese Long Short-Term Memory (LSTM) to teach our system that two documents are similar or not, and finally, use the CNN technique to identify the various types of plagiarism.

III. METHODOLOGY

In order to carry out the proposed approach for detection of plagiarism, the following paragraphs will explain some required methodologies.

A. Text Parsing and Preprocessing

A Text Parsing is a process of extracting text from documents such as Docx and PDF files. Convert text in a certain format into a more usable format [15].

In the proposed approach, after extracting the text, the parser will extract the title and author name of the suspicious paper. Utilizing Regex and string manipulations, the unnecessary text can be deleted. The text segments before the abstract heading and after the acknowledgement or reference headers are deleted from the suspicious document's text so that the deleted texts are not tested for plagiarism.

NLP is a wide term that covers a wide range of activities and applications, such as text categorization, machine translation, text similarity detection, and so on [16], [17].

Pre-processing a text simply means converting it into a format that the machine can interpret, predict, and analyze using various machine learning methods utilizing the NLP techniques [16], [18].

The following are some of the most commonly utilized pre-processing techniques utilizing NLP [8], [19]:

•Lowercase the words: since the computer assumes that the same words with one capital and one lowercase are two different words.

•Tokenization: the process of breaking down sentences into words, letters, and punctuations, this stage aids in the subsequent processing processes by filtering out undesirable terms.

•Stop words Removal: Words like "the," "are," "is," "and," and so on have no meaning and these extra words aren't given any weight.

•Remove all punctuation marks because the machine doesn't understand punctuation and its presence causes the text to become loud.

•Stemming: is a word shortening strategy that seeks to get a word back to its root, but the semantic meaning of all the forms remains the same.

•Lemmatization: is the process of deleting or changing a suffix from a word in order to return it to its original form.

The process of generating a set of n-tokens of a text from a text document or corpus is known as N-gram. The n-gram can do at character level, word level or even at phrase level [20], [21]. This proposed approach utilized splitting the text into n-words.

B. The Corpus

In a PD system, a suspicious document is compared to a corpus or set of reference documents [15]. It is possible that this reference collection will be available online or offline, using online sources on the internet or an offline database with the source materials [22].

The Turnitin platform is a public used web site for detection of plagiarism and similarity index, it has a huge online and off-line corpus of text documents [23].

To create the corpus and since it is impossible to have a huge corpus of documents as that of for example, the Turnitin platform, the proposed approach created a corpus consisting of 500 scientific papers as text files, that files are created from the text of pdf or docx files, the text files are parsed and preprocessed then stored in the corpus. Furthermore, A Comma Separated Values (CSV) file containing the title, author, keywords, count of pages, count of sentences, count of words, and size of each scientific publication has been prepared.

C. The Clustering of document

Clustering is an intelligent technique that groups objects together according to common features. Document clustering Fig. 1, often known as text grouping, is primarily concerned with grouping documents based on comparable content or file categories. NLP techniques can be used to identify similarity between documents based on semantic relationships [12].

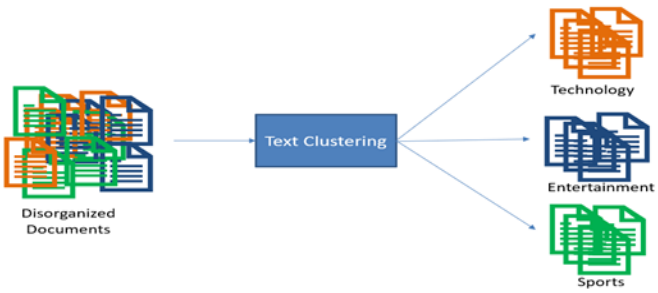

Fig. 1. Document Clustering

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The Mini-Batch K-means algorithm for clustering is an improved version of k-means algorithm, the most widely used technique for clustering documents. The algorithm of Mini-Batch K-means is both simple and quick [24]. The Mini-Batch K-mean was utilized in certain research publications because it can work with large datasets and enhance accuracy [12], [17]. In the proposed approach, the documents are subsequently grouped using the Mini-Batch K-mean approach according to titles and keywords of the scientific documents.

D. Extraction of Textual features

Textual feature extraction also known as text encoding, vector representations and word or sentence embedding. It is the process to extract numeric features of text in order for machines to understand them [18], [22]. The sentence embedding involves encoding the text into large vectors which preserves the contextual and semantic meaning of sentences, these vectors can be utilized for a variety of purposes, including text categorization, grouping, semantic similarity, etc.[5], [25].

This paper used USE which is a pretrained deep learning-based feature extraction model. The two USE variants are the Transformer encoder and Deep Averaging Network (DAN). The first one is more accurate, but it costs a lot to compute. The second one is less computationally demanding but less precise.

When English text of varying length is input, the result is a 512-dimensional vector.

A DAN encoder is utilized to train the used model. The model generates embeddings from a sentence or word as input as illustrated in Fig. 2. They function as follows [14], [25]:

a) Sentence tokenization is carried out after lower case conversion.
b) The average of the word and bi-gram embeddings within a sentence is calculated.
c) Pass it through a single or more layers feed-forward deep neural network.
d) The sentence is then converted into a 512-dimensional embedding. During training, the embeddings for words and bi-grams are discovered.

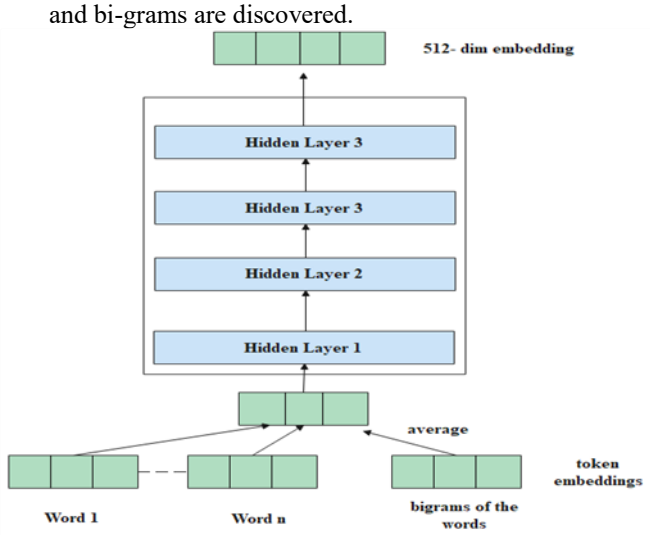

Fig. 2. The architecture of the Deep Average Network [14], [25]

For the purpose of creating 512 dimensional final embeddings for sentences, the DAN computes unigram and bigram embeddings before passing the results to a deep neural network

E. Text Similarity Measurement Technique

Following the acquisition of each sentence's vector representation, the similarity measure can be used to detect similarity between each two-sentence pair [26], [27].

The cosine similarity method determines the direction of two vectors by measuring the cosine angle between them[28]. The similarity between two vectors is calculated using the ratio between their dot products and the product of their lengths [28], [29].

If Su is vector of the suspicious text and So is the vector of the source text, then the cosine similarity between Su and So is:

$$Cos(Su, So) = \frac{Su.So}{\|Su\| \times \|So\|} = \frac{\sum_{i=1}^{n} Su_i \times So_i}{\left(\sqrt{\sum_{i=1}^{n} So_i^2} \times \sqrt{\sum_{i=1}^{n} So_i^2}\right)} \quad (1)$$

Because it does not rely on other distance measurements, cosine similarity works well even when the data is dispersed across multiple dimensions[28], [30].

IV. THE PROPOSED APPROACH

Using the methodologies provided in the methodology section, the proposed approach for semantic PD utilized USE and cosine similarity measure. In approach has several stages. The proposed approach is explained in the next sections.

As shown in Fig. 3, the Semantic PD approach is carried out through several steps as following:

1. To create the corpus, 500 scientific papers as pdf and docx files are collected, then the text from each file is extracted and saved as a txt file. The topics of the files are divided into six categories: computer science, arts, physics and chemistry, geography and history, biology, and sports. Table I shows the number of papers in each category.

TABLE I. TOPICS AND COUNT OF SCIENTIFIC PAPERS

Topic	Count of scientific papers
Computer science	106
Arts	76
Physics and Chemistry	85
Geography and History	93
Biology	86
Sports	54
Total	500

2. The suspicious file is uploaded by user and N value is selected for splitting text into n-word tokens and a threshold value.
3. The keywords, Title, and other information are extracted from the suspicious document, Title and Keywords are encoded using USE model.
4. Parse the suspicious document and extract the text between abstract and acknowledge or references headings.

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5. Preprocess the parsed document utilizing the NLP and perform Mini-Batch K-means clustering on corpus text to get the related documents according to the numerical vectors of title and keywords.
6. Split the text of source documents in corpus into n-words.
7. Segment the suspicious text into segments of heading sections and their paragraphs, like Abstract, Introduction, etc.., the texts of Abstract, Result and conclusion heading sections will have high weights and compared with threshold – 0.1 to detect plagiarism. Since the author must not plagiarize any text at these segments.
8. For the paragraphs of each section of a suspicious document, apply n-gram to convert it to n-word tokens.
9. Calculate the USE text encoding.
10. Apply the cosine similarity between suspicious and corpus text vectors. Each similarity greater than threshold is marked as plagiarized.
11. Generate the detailed report that contains the total percentage of plagiarism and percentage of plagiarism from each source. Highlight the plagiarized sentences according to assigned color for the source of each sentence and other information

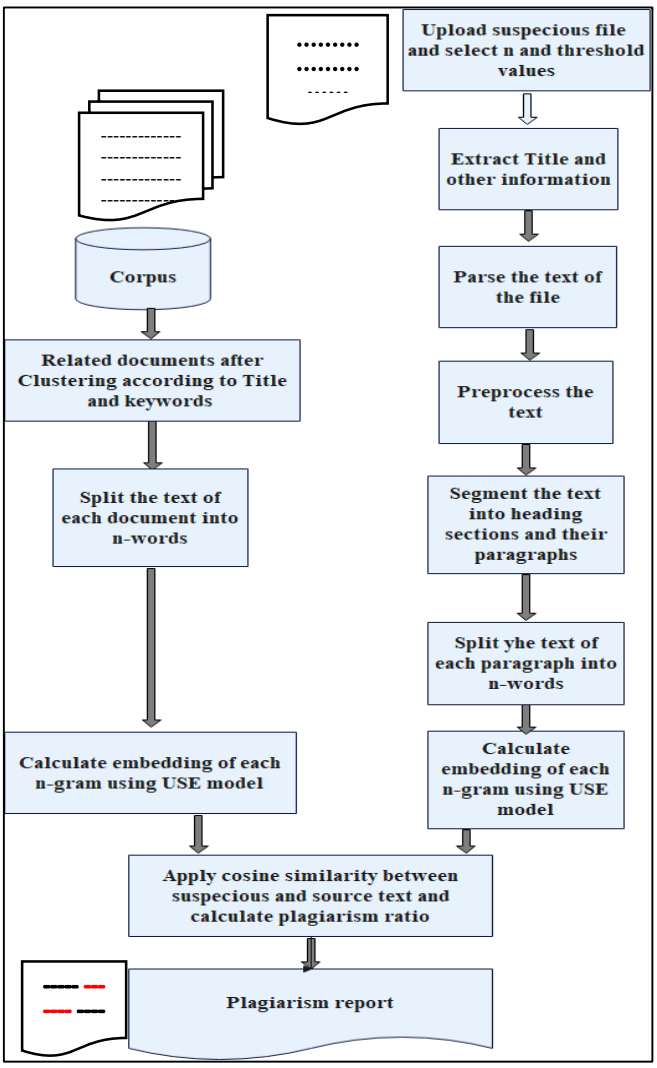

Fig. 3. The Proposed Semantic PD

V. RESULTS

In this paper, the phases of the suggested method were applied to a number of suspicious documents. Regardless of whether the document is a pdf or a docx, it will be converted to text and processed according to the rules of the proposed system.

The results of Semantic PD of ten suspicious documents are shown in Table II. as percentages of plagiarism, when n-words equal to 5 and 7.

The application will decide whether to accept or reject the suspicious document based on a threshold value, such as that used in many scientific institutes. The document is accepted if the percentage of plagiarism is less than 20%; else, the document is rejected and must be re-edited before being checked again.

TABLE II. PERCENTAGES OF SEMANTIC PLAGIARISM.

Doc No.	Semantic PD Percentage			
	N = 5	Accept	N = 7	Accept
1	0.26	No	0.23	No
2	0.13	Yes	0.11	Yes
3	0.25	No	0.21	No
4	0.22	No	0.18	Yes
5	0.21	No	0.18	Yes
6	0.17	Yes	0.12	Yes
7	0.23	No	0.20	No
8	0.27	No	0.22	No
9	0.46	No	0.41	No
10	0.18	Yes	0.15	Yes

An easy-to-use user interface has been used to present the system, it accepts the input Pdf, Docx or txt file formats. The file is uploaded to the system. After that, it is verified for plagiarism.

The result is displayed as a plagiarism report which indicates the acceptance of the document according to plagiarism ratio. The ratio can be less than 20% and is accepted and rejected if the ratio is exceeded 20 %.

To assess if the proposed method is reliable or not, a sample of five suspicious documents are checked for plagiarism using Turnitin web platform, then collect the source documents for each suspicious document. The collected documents are considered as source documents in the proposed system and each of the suspicious documents are checked for plagiarism.

Table III illustrates the percentages of plagiarism of each suspicious document in Turnitin and proposed system without any filters and with excluding bibliography and quotes and the time required for each document as minutes and seconds.

Figure 4. shows the percentages of plagiarism of each suspicious document in Turnitin and proposed system with and without filtering.

TABLE III. THE PERCENTAGES OF PLAGIARISM AND RUN TIME OF EACH SUSPICIOUS DOCUMENT IN TURNITIN AND PROPOSED SYSTEM

Doc No.	Filters	Turnitin		The Proposed System	
		Percentage	Time M:S	Percentage	Time M:S
1	No filter	0.93	7:37	0.92	13:32

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	Excluding bibliography and quotes	0.91	7:37	0.91	13:27
2	No filter	0.17	6:49	0.18	11:53
	Excluding bibliography and quotes	0.15	6:49	0.16	11:47
3	No filter	0.26	5:52	0.27	12:17
	Excluding bibliography and quotes	0.25	5:52	0.25	12:12
4	No filter	0.32	6:54	0.31	13:24
	Excluding bibliography and quotes	0.32	6:54	0.30	13:21
5	No filter	0.28	5:42	0.28	14:13
	Excluding bibliography and quotes	0.27	5:42	0.26	14:09

After carrying out the experiments and comparing with the results of the Turnitin platform, the proposed system is reliable and produced reasonable and glorious results regarding the accuracy. On the other hand, The Turnitin platform is better regarding the execution time since the proposed approach is using pretrained USE sentence embedding model which requires more computational resources and thus execution time. However, the USE model provides quick results and require fewer resources compared with Bidirectional Encoder Representations from Transformers (BERT) and Embeddings from Language Models (ELMo) pretrained models.

Generally, since the speed of the internet is not stable and the loads on CPU which affect to the execution time, the accuracy is more important than the speed of processing in such plagiarism checking systems and the majority of authors make attention to accuracy rather than the run time.

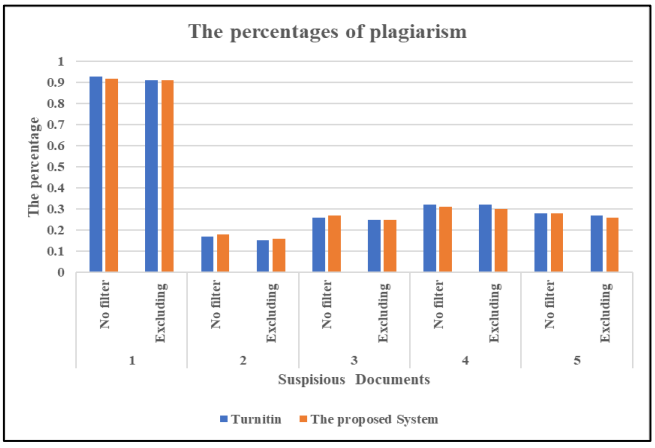

Fig. 4. The percentages of plagiarism of each suspicious document in Turnitin and proposed system.

VI. Discussion

The steps of the proposed system in this paper were carried out on many of the source and suspicious documents utilizing the Python language with its libraries such as Natural language Tokit (NLTK), regular expressions (re) and others in the Flask framework. Whatever sort of document it is, it will be converted to text and processed according to the system's steps. The information of each scientific paper including title, author, keywords, year of publishing, topic, size, count of words, count of sentences and number of pages are stored in a CSV file while the corpus contains 500 text files of the scientific papers.

In this paper, work was based on intelligent USE model for detection of semantic plagiarism. The input is document file and the output is a report about the semantic plagiarism that exists in the suspicious document after checking against the source documents in the corpus. According to the experiments, semantic plagiarism has been detected in a sample of documents stored in the corpus with different percentages. The value of n is very important for text segmentation, after many experiments its noticed that when this value increases the plagiarism percentage is decreases and vice versa.

VII. Conclusions and Suggestions

A plagiarism detector has been developed in this paper employing machine learning algorithms such as USE for text encoding, Mini-Batch K-means for clustering, NLP techniques and cosine similarity. The technology works well and detects the level of plagiarism in the text files provided. This technique may simply be used in educational institutions such as schools and colleges to detect plagiarism. The system's operation does not necessitate any complicated instructions or training. It's a plagiarism detection method that's quick, simple, and effective.

The findings of this paper may make it easier to spot similarities in each scientific paper and determining whether or not the proposed paper fits the originality semantically. However, this method is limited since it only uses few counts of documents in the corpus. More investigation may be done to see whether a corpus that is more compact and has a more and more data might be used.

This paper advises adding techniques to increase system performance and accuracy such as assign similarity weights for each header and its section in the suspicious paper, increase the count of documents in the corpus in the current used topics and from many other topics and utilizing the intelligent clustering techniques in selection of the related source documents from the corpus.

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